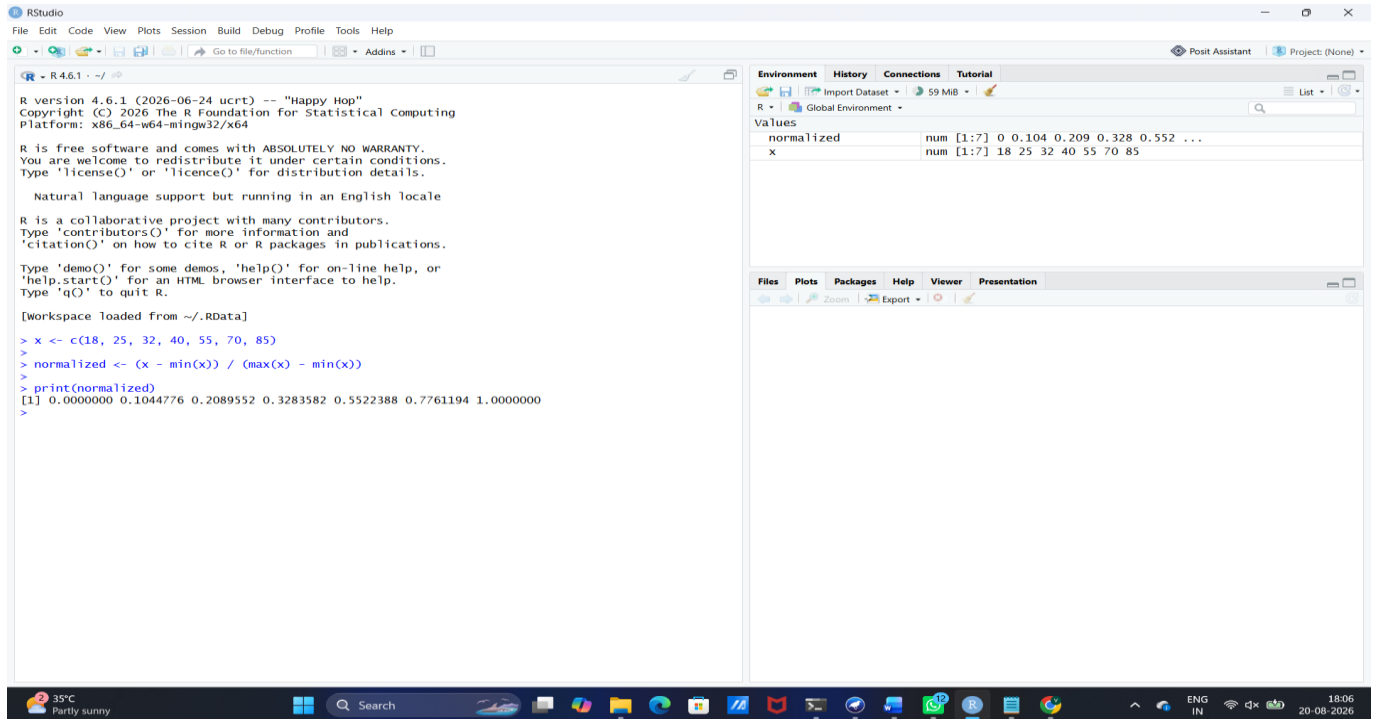


## SET 3

1. R: Perform min-max normalization on 18,25,32,40,55,70,85 to range 0–1.



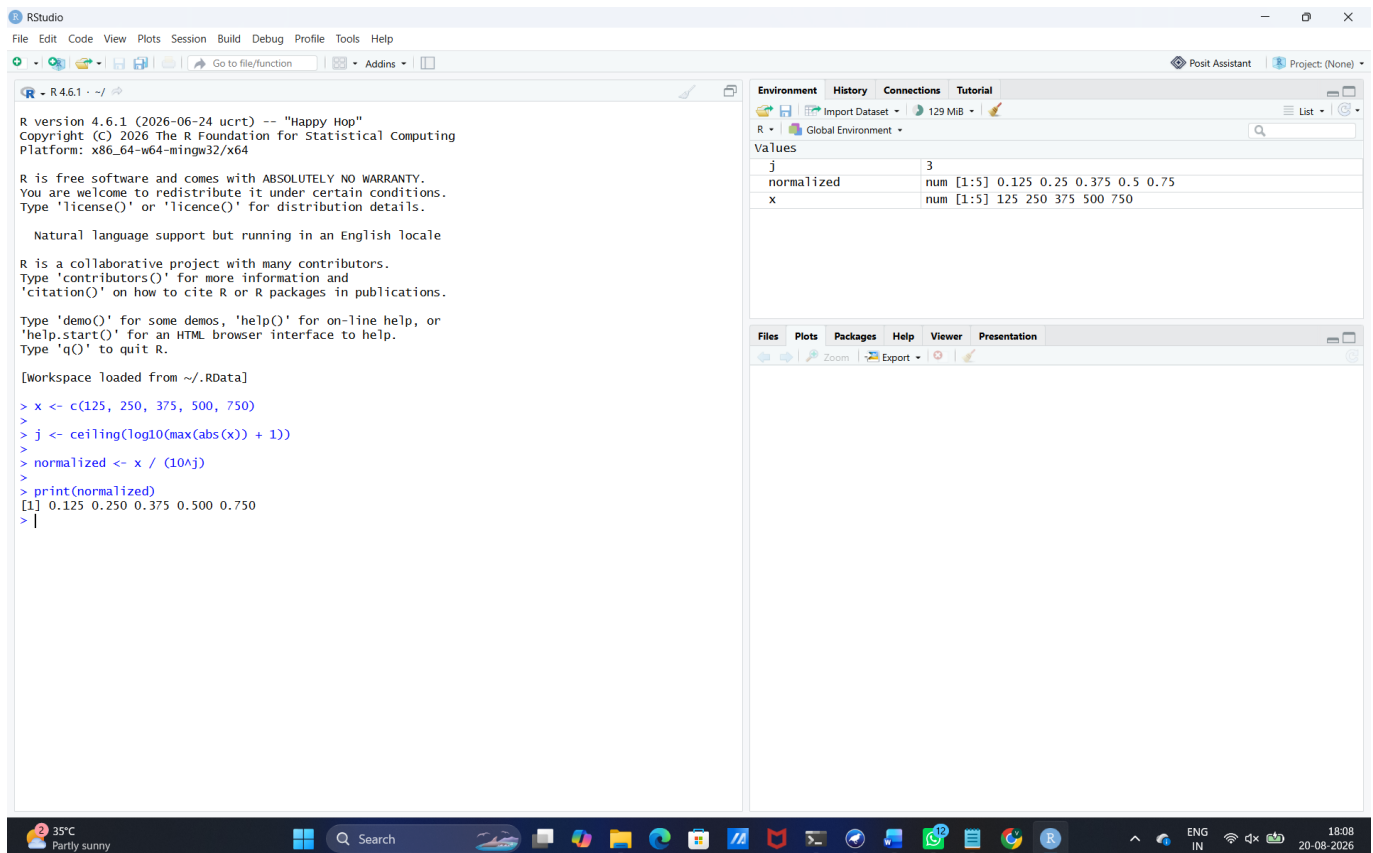
The screenshot shows the RStudio interface. The console on the left displays the R startup message and the execution of the following code:

```
[Workspace loaded from ~/.RData]
> x <- c(18, 25, 32, 40, 55, 70, 85)
>
> normalized <- (x - min(x)) / (max(x) - min(x))
>
> print(normalized)
[1] 0.0000000 0.1044776 0.2089552 0.3283582 0.5522388 0.7761194 1.0000000
>
```

The Environment pane on the right shows the following values:

| Variable   | Class     | Value                         |
|------------|-----------|-------------------------------|
| normalized | num [1:7] | 0 0.104 0.209 0.328 0.552 ... |
| x          | num [1:7] | 18 25 32 40 55 70 85          |

2. R: Perform decimal-scaling normalization on 125,250,375,500,750.



The screenshot shows the RStudio interface. The console on the left displays the R startup message and the execution of the following code:

```
[Workspace loaded from ~/.RData]
> x <- c(125, 250, 375, 500, 750)
>
> j <- ceiling(log10(max(abs(x)) + 1))
>
> normalized <- x / (10^j)
>
> print(normalized)
[1] 0.125 0.250 0.375 0.500 0.750
> |
```

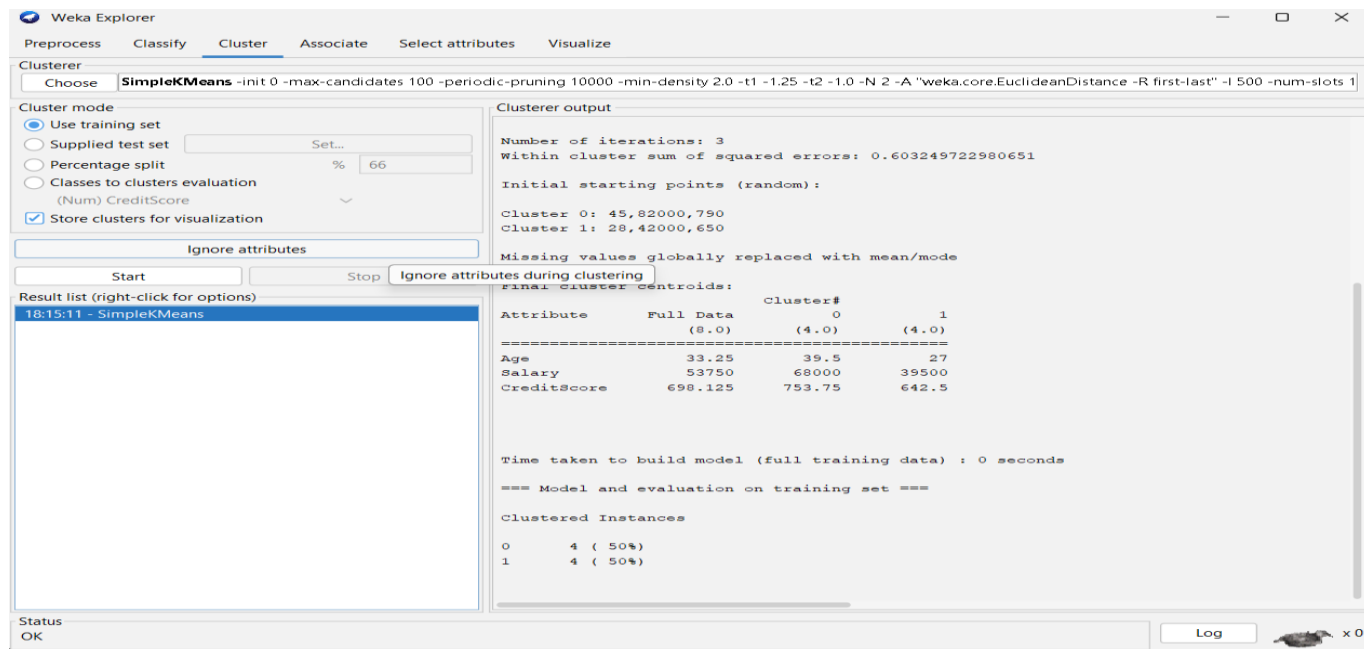
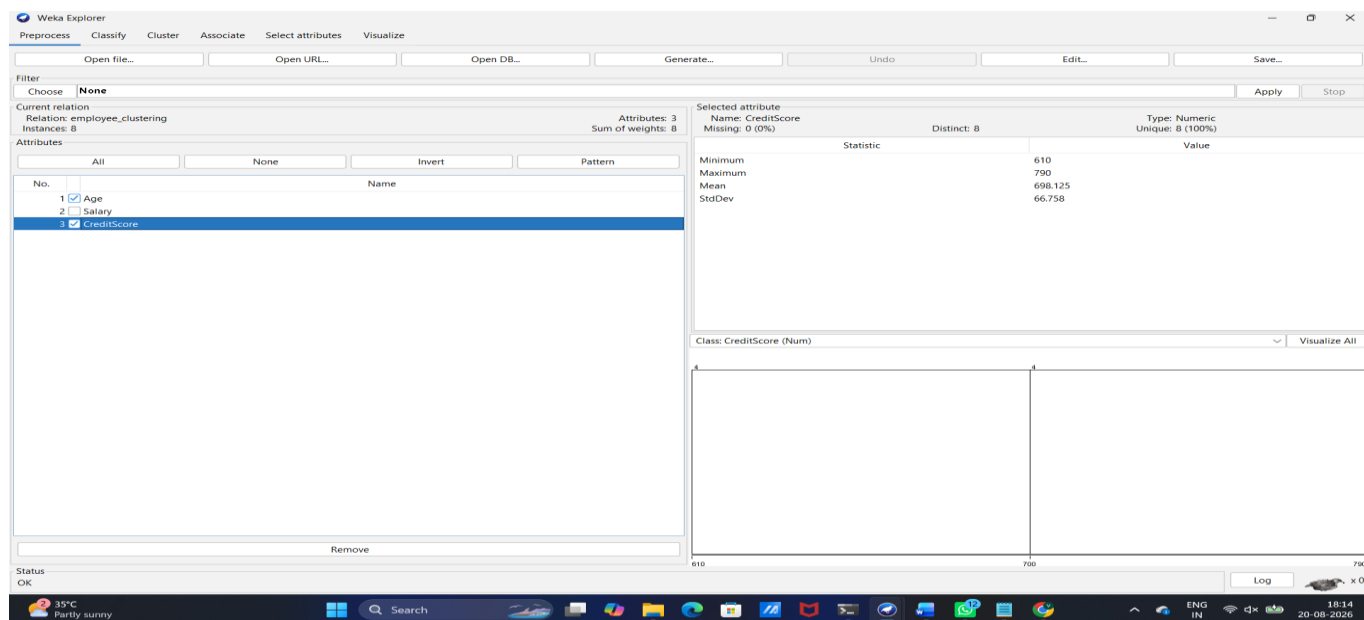
The Environment pane on the right shows the following values:

| Variable   | Class     | Value                     |
|------------|-----------|---------------------------|
| j          | 3         |                           |
| normalized | num [1:5] | 0.125 0.25 0.375 0.5 0.75 |
| x          | num [1:5] | 125 250 375 500 750       |

3. WEKA: Create employee Age, Salary, CreditScore ARFF data; apply SimpleKMeans with k=2 and visualize clusters.

### Dataset: Employee Clustering

```
@relation employee_clustering
@attribute Age numeric
@attribute Salary numeric
@attribute CreditScore numeric
@data
25,35000,620
28,42000,650
31,48000,690
35,55000,720
40,70000,760
45,82000,790
24,33000,610
38,65000,745
```



4. WEKA: Repeat employee clustering with  $k=3$  and compare cluster sizes, centroids and within-cluster error.

Dataset: Employee Clustering –  $k=3$

Use the same Employee Clustering ARFF dataset from Set 3 Q3.

The screenshot shows the Weka Explorer application with the 'Cluster' tab selected. The 'SimpleKMeans' algorithm is chosen, and the 'Use training set' option is selected for the cluster mode. The 'Store clusters for visualization' checkbox is checked. The 'Clusterer output' pane displays the following results:

```
test mode: evaluate on training data

=== Clustering model (full training set) ===

kMeans
=====

Number of iterations: 2
Within cluster sum of squared errors: 0.3007778823295506

Initial starting points (random):

Cluster 0: 45,82000,790
Cluster 1: 28,42000,650
Cluster 2: 35,55000,720

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute      Full Data      Cluster#
              (8.0)         (2.0)         (3.0)         (3.0)
=====
Age             33.25          42.5          25.6667        34.6667
Salary          53750          76000         36666.6667      56000
CreditScore     698.125        775           626.6667       718.3333

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      2 ( 25%)
1      3 ( 38%)
2      3 ( 38%)
```

The status bar at the bottom indicates 'Status OK' and the system clock shows 18:20 on 20-08-2026.